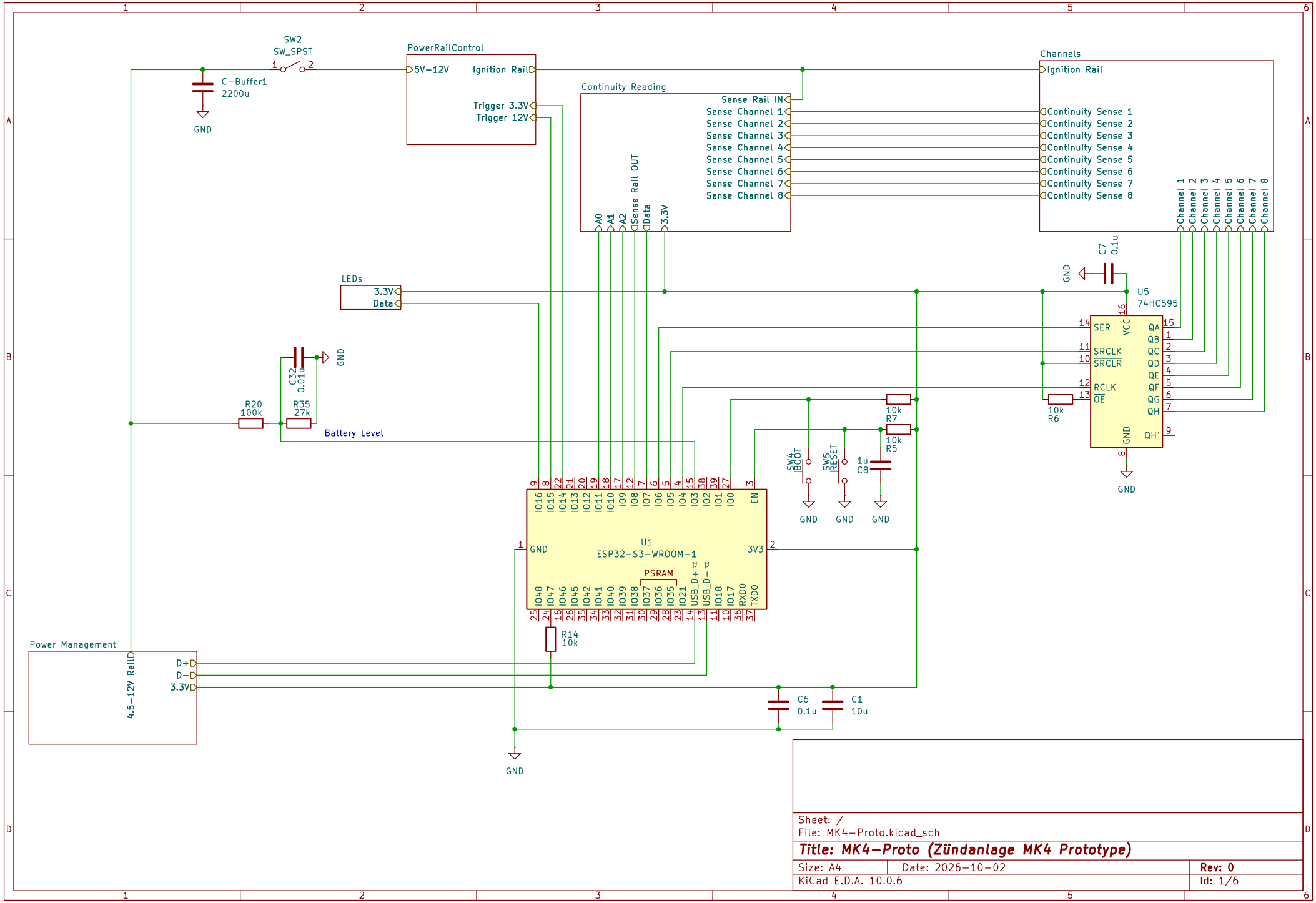
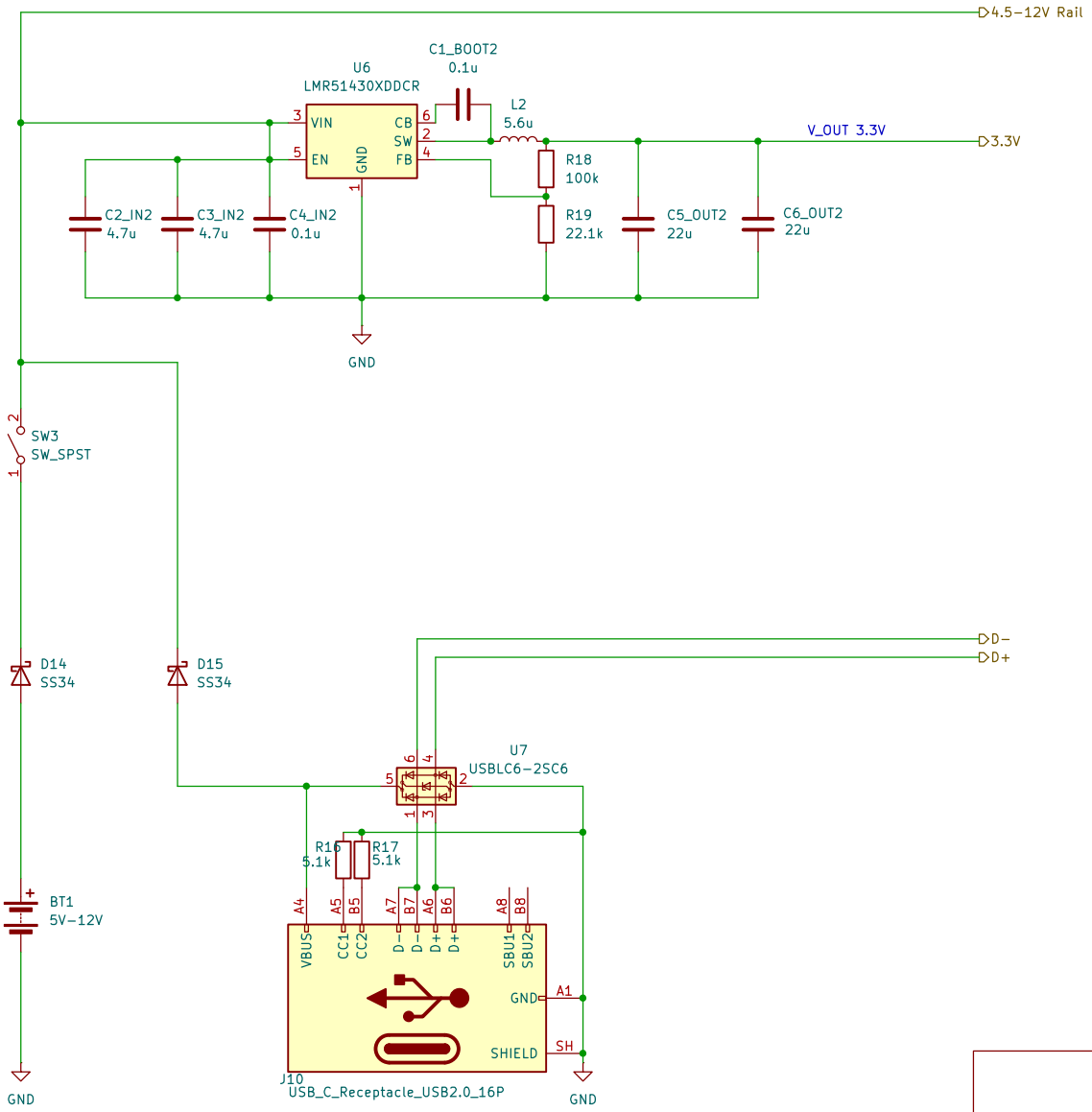
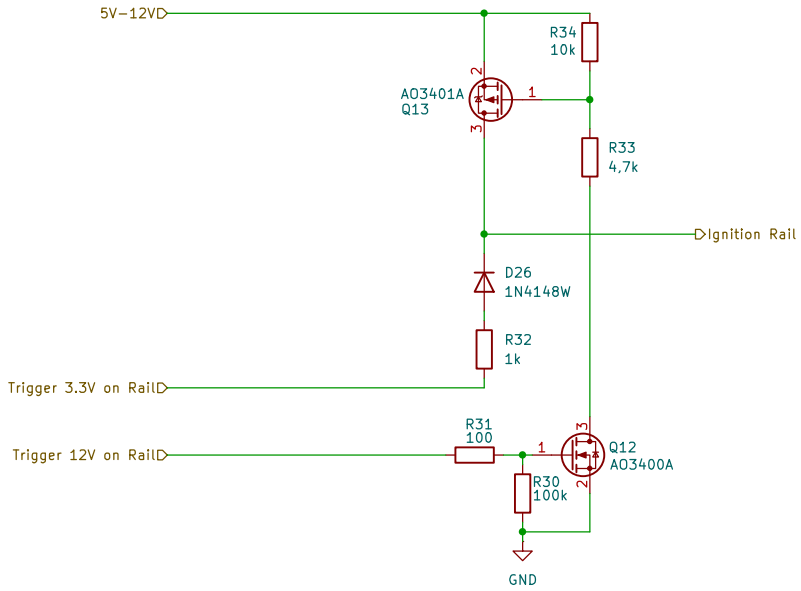
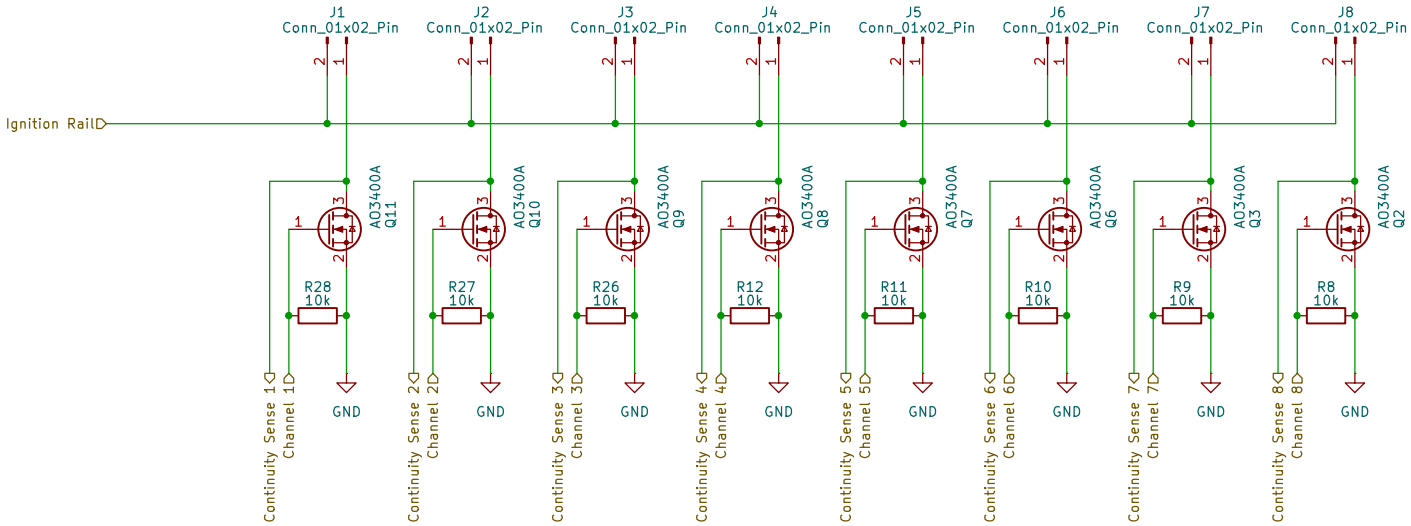

SW2
SW_SPST
C-Buffer1
2200u
GND
PowerRailControl
5V-12V
Ignition Rail
Trigger 3.3V
Trigger 12V
Continuity Reading
Sense Rail IN
Sense Channel 1
Sense Channel 2
Sense Channel 3
Sense Channel 4
Sense Channel 5
Sense Channel 6
Sense Channel 7
Sense Channel 8
A0
A1
A2
Sense Rail OUT
Data
3.3V
Channels
Ignition Rail
Continuity Sense 1
Continuity Sense 2
Continuity Sense 3
Continuity Sense 4
Continuity Sense 5
Continuity Sense 6
Continuity Sense 7
Continuity Sense 8
Channel 1
Channel 2
Channel 3
Channel 4
Channel 5
Channel 6
Channel 7
Channel 8
C7
0.1u
GND
LEDs
3.3V
Data
U5
74HC595
SER
SRCLK
SRCLR
RCLK
OE
VCC
GND
QA
QB
QC
QD
QE
QF
QG
QH
QH'
R6
10k
R7
10k
R5
10k
C32
0.01u
R20
100k
R35
27k
Battery Level
SW4
BOOT
SW5
RESET
C8
1u
U1
ESP32-S3-WROOM-1
GND
3V3
EN
PSRAM
USB_D+
USB_D-
RXD0
TXD0
R14
10k
Power Management
4.5-12V Rail
D+
D-
3.3V
C6
0.1u
C1
10u
Sheet: /
File: MK4-Proto.kicad_sch
Title: MK4-Proto (Zündanlage MK4 Prototype)
Size: A4
Date: 2026-10-02
Rev: 0
KiCad E.D.A. 10.0.6
Id: 1/6



Sheet: /Power Management/		
File: PowerManagement.kicad_sch		
Title: Power Management		
Size: A4	Date: 2026–10–24	**Rev:**
KiCad E.D.A. 10.0.6		Id: 2/6



Sheet: /PowerRailControl/		
File: testing.kicad_sch		
Title: Ignition & Test Rail Control		
Size: A4	Date: 2026–10–24	**Rev:**
KiCad E.D.A. 10.0.6		Id: 3/6

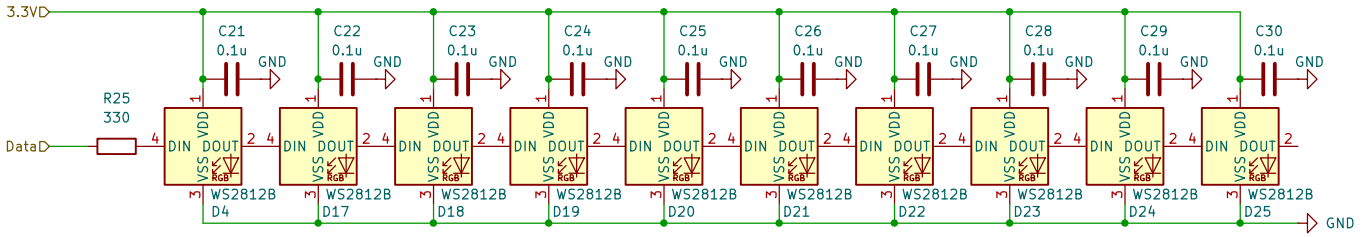


Current measurement line – Continuity Sense
Channel On/Off control by Mosfets
One ingition current rail

Sheet: /Channels/
File: channels.kicad_sch

Title: Channels

Size: A4	Date: 2026-10-25	**Rev:**
KiCad E.D.A. 10.0.6		Id: 4/6

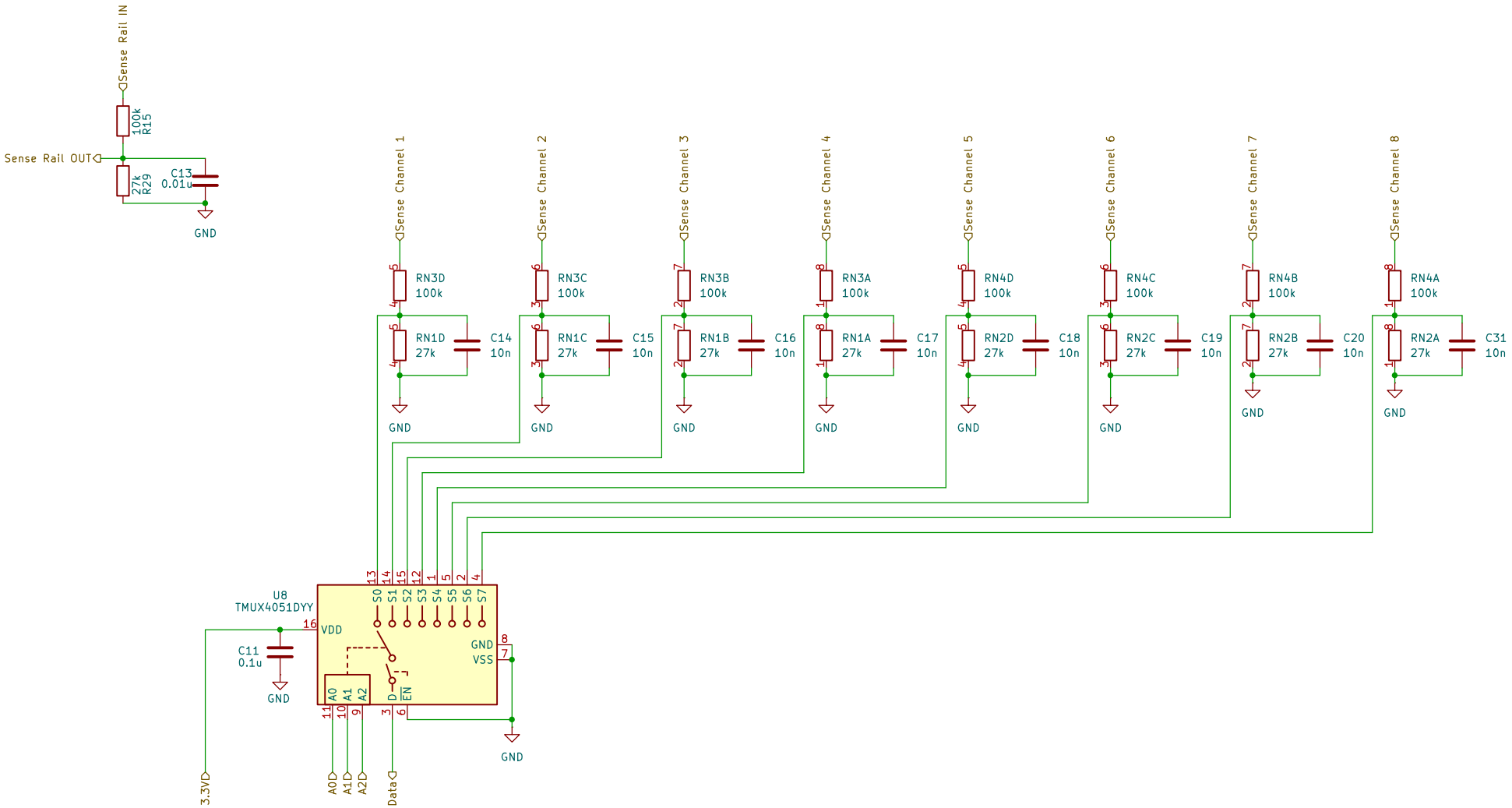


2 Status Leds
8 Channel Leds

Sheet: /LEDs/
File: leds.kicad_sch

Title: LEDs

Size: A4	Date: 2026-10-24	**Rev:**
KiCad E.D.A. 10.0.6		Id: 5/6



TMUX4051 8:1 mux, A0-A2 select, D -> ADC, always enabled
8-CH VOLTAGE SENSE: 100k/27k divider (0-15.5 V -> 0-3.3 V), 10 nF filter (fc ≈ 750 Hz)

Sheet: /Continuity Reading/
File: ContinuityReading.kicad_sch

Title: Continuity Sensing

Size: A4	Date:	Rev:
KiCad E.D.A. 10.0.6		Id: 6/6